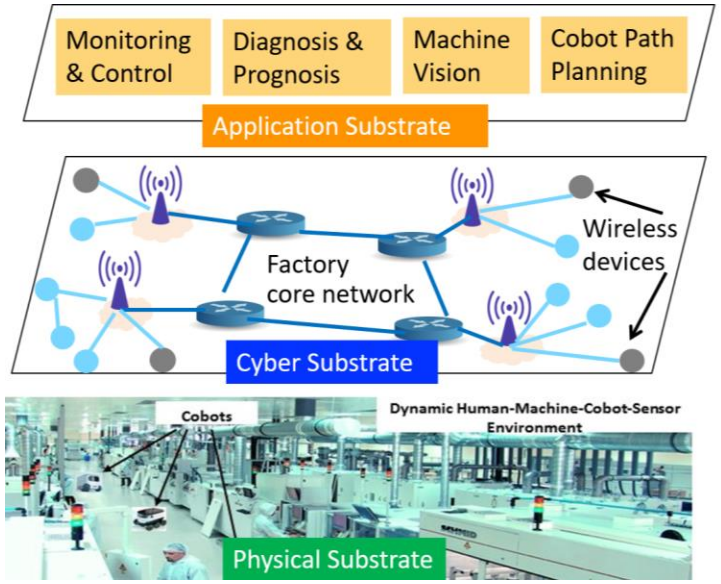
Track 1: Collaborative Research: VINES: Track 1: NSF-SRC: Towards Resilient and Converged NextG Manufacturing Systems (2549126)

Problem to Solve

A central challenge in NextG manufacturing system (NGMS) is ensuring resilient and deterministic connectivity across wired and wireless domains in dynamic, failure-prone environments.

Existing solutions are largely reactive and siloed, leaving manufacturing systems vulnerable to interference, equipment failures, and unpredictable disruptions.



Vector A

- Resilience: maintain operation despite interference or faults
- Convergence: coordinate across wired/wireless domains with bounded latency and predictable performance

Vector B

- AI-Native Design: ML-driven, short-horizon prediction at the network edge to support timely and reliable adaptation.
- ISAC: Incorporate sensing and communication information into network awareness and coordination.

Convergence

The enabling technologies supply the information and decision support to realize the NGMS capabilities through short-horizon prediction, proactive edge/core adaptation and contract-based end-to-end integration.

Project Contributions

Advance the cyber infrastructure design for NGMS by enabling predictive coordination across converged wireless and wired networks, establishing a resilient communications foundation.

Our Approach

- Enhance wireless edge with ML-driven perception to adapt scheduling via ISAC-based environmental awareness.
- Extend the TSN core with hybrid hardware-software mechanisms to preserve determinism under faults.
- Establish a cross-domain control layer to unify predictive wireless edge and deterministic core into verifiable end-to-end guarantees.

Lead: U of Connecticut

Collaborative: U of Iowa

Collaborative: MIUN